

Week 5

- **Problem 1**

Let $A, B \in M_n(\mathbb{R})$ satisfy

$$AB = A \quad \text{and} \quad BA = B.$$

1. Prove that

$$A^2 = A \quad \text{and} \quad B^2 = B.$$

2. Does it necessarily follow that $A = B$? Prove your answer. If the statement is false, give explicit matrices

$$A, B \in M_2(\mathbb{R})$$

satisfying the two given equations but with $A \neq B$.

Your argument should use only the arithmetic properties of matrices developed in the section.

- **Solution 1.1**

1. Suppose $A, B \in M_n(\mathbb{R})$ satisfy $AB = A$, $BA = B$. By using the associative law and substituting the given equations, we have

$$A^2 = AA = (AB)A = A(BA) = AB = A.$$

$$B^2 = BB = (BA)B = B(AB) = BA = B.$$

2. No, it does not necessarily follow that $A = B$. Let the two non-equal 2×2 matrices be defined as

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

We verify their algebraic products directly

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 1 & 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 1 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = A.$$

$$BA = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = B.$$

Since $A \neq B$, the statement is false.

- **Problem 2**

Write the polynomials

1. $f(x) = x^6 - 5x^4 - 4x^2 + 20$

2. $g(x) = 2x^5 - 2x^4 - 2x^3 + 2x^2 - 4x + 4$

as a product of irreducible factors over $\mathbb{Z}[x]$, $\mathbb{Q}[x]$, $\mathbb{R}[x]$, and $\mathbb{C}[x]$.

- **Solution 2.1**

1. We factor $f(x)$ by grouping the terms systematically

$$f(x) = x^4(x^2 - 5) - 4(x^2 - 5) = (x^4 - 4)(x^2 - 5) = (x^2 - 2)(x^2 + 2)(x^2 - 5).$$

By definition, we have

■ Over $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$

$$f(x) = (x^2 - 2)(x^2 + 2)(x^2 - 5).$$

- Over $\mathbb{R}[x]$

$$f(x) = (x - \sqrt{2})(x + \sqrt{2})(x^2 + 2)(x - \sqrt{5})(x + \sqrt{5}).$$

- Over $\mathbb{C}[x]$

$$f(x) = (x - \sqrt{2})(x + \sqrt{2})(x - \sqrt{2}i)(x + \sqrt{2}i)(x - \sqrt{5})(x + \sqrt{5}).$$

2. We factor $g(x)$ by grouping the terms systematically

$$\begin{aligned} g(x) &= 2x^4(x - 1) - 2x^2(x - 1) - 4(x - 1) \\ &= 2(x - 1)(x^4 - x^2 - 2) \\ &= 2(x - 1)(x^2 - 2)(x^2 + 1). \end{aligned}$$

By definition, we have

- Over $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$

$$g(x) = 2(x - 1)(x^2 - 2)(x^2 + 1).$$

- Over $\mathbb{R}[x]$

$$g(x) = 2(x - 1)(x - \sqrt{2})(x + \sqrt{2})(x^2 + 1).$$

- Over $\mathbb{C}[x]$

$$g(x) = 2(x - 1)(x - \sqrt{2})(x + \sqrt{2})(x - i)(x + i).$$

• Problem 3

Let $n \in \mathbb{N}$ such that n is even. Show that the polynomial $p(x) = x^n + x^{n-1} + \dots + x + 1$ has no rational roots.

◦ Proof 3.1

By contradiction. Suppose $p(x)$ has a rational root $\alpha = \frac{r}{s} \in \mathbb{Q}$ where $r, s \in \mathbb{Z}$ are coprime. By the Rational Roots Theorem, since the leading coefficient is 1 and the constant term is 1, we must have

$$r \mid 1 \implies r = \pm 1 \quad \text{and} \quad s \mid 1 \implies s = \pm 1.$$

Thus, the only possible rational roots are $\alpha = 1$ or $\alpha = -1$. We evaluate $p(x)$ at these two points

- For $\alpha = 1$

$$p(1) = \underbrace{1^n + 1^{n-1} + \dots + 1 + 1}_{n+1 \text{ terms}} = n + 1.$$

Since $n \in \mathbb{N}$, $n + 1 \geq 2 \neq 0$, so 1 is not a root.

- For $\alpha = -1$

$$p(-1) = (-1)^n + (-1)^{n-1} + \dots + (-1) + 1.$$

Since n is even, the exponents alternate between even and odd integers, producing a geometric series of alternating signs

$$p(-1) = 1 - 1 + 1 - 1 + \dots - 1 + 1 = 1.$$

Since $p(-1) = 1 \neq 0$, -1 is not a root.

Therefore, $p(x)$ has no rational roots.

• Problem 4

Let $a, b \in \mathbb{K}$ with $a \neq b$, and let $p(x) \in \mathbb{K}[x]$. Find the remainder when $p(x)$ is divided by $(x - a)(x - b)$ in terms of $p(a)$ and $p(b)$.

◦ **Solution 4.1**

By the division algorithm for polynomials, there exist unique polynomials $q(x), r(x) \in \mathbb{K}[x]$ such that

$$p(x) = (x - a)(x - b)q(x) + r(x)$$

with either $r(x) = 0$ or $\deg(r(x)) < \deg((x - a)(x - b)) = 2$. This implies $\deg(r(x)) \leq 1$, so we can write the remainder as $r(x) = Bx + C$ for some constants $B, C \in \mathbb{K}$.

We evaluate the equation $p(x) = (x - a)(x - b)q(x) + Bx + C$ at the roots $x = a$ and $x = b$

$$\begin{aligned} p(a) &= (a - a)(a - b)q(a) + Ba + C = Ba + C. \\ p(b) &= (b - a)(b - b)q(b) + Bb + C = Bb + C. \end{aligned}$$

Then we solve this linear system of equations for B and C . We have

$$\begin{cases} B = \frac{p(a) - p(b)}{a - b} \\ C = \frac{ap(b) - bp(a)}{a - b} \end{cases}.$$

Substituting the expressions for B and C back into $r(x) = Bx + C$ yields

$$r(x) = \left(\frac{p(a) - p(b)}{a - b} \right) x + \frac{ap(b) - bp(a)}{a - b} = \frac{p(a)(x - b) - p(b)(x - a)}{a - b}.$$

Therefore, the required remainder is $\frac{p(a)(x - b) - p(b)(x - a)}{a - b}$.

• **Problem 5**

Let

$$A = (a_{ij}) \in M_{m \times n}(\mathbb{R}).$$

Prove that the following three statements are equivalent

$$A = 0, \quad A^\top A = 0_{n \times n}, \quad AA^\top = 0_{m \times m}.$$

By saying that the three statements are equivalent, you must prove that each one holds if and only if the others hold. In particular, it is sufficient to establish

$$\begin{aligned} A = 0 &\implies A^\top A = 0, \\ A^\top A = 0 &\implies A = 0, \end{aligned}$$

and

$$\begin{aligned} A = 0 &\implies AA^\top = 0, \\ AA^\top = 0 &\implies A = 0. \end{aligned}$$

Your proof should use only the definitions of matrix multiplication and transpose. In particular, when proving

$$A^\top A = 0 \implies A = 0,$$

examine the diagonal entries of $A^\top A$ and justify carefully why they force every entry of A to be zero. Similarly, for

$$AA^\top = 0 \implies A = 0,$$

examine the diagonal entries of $AA^\top$.

◦ **Proof 5.1**

Let $A = [a_{ij}]_{m \times n}$, which means $A^\top = [a_{ji}]_{n \times m}$.

1. **Proof of $A = \mathbf{0} \implies A^\top A = \mathbf{0}$**

$$A = \mathbf{0} \implies a_{ij} = 0 \quad (\forall 1 \leq i \leq m, 1 \leq j \leq n).$$

The (k, j) -th entry of $A^\top A$ is given by

$$\sum_{i=1}^m a_{ik} a_{ij} = \sum_{i=1}^m 0 \cdot 0 = 0 \quad (\forall 1 \leq k, j \leq n) \implies A^\top A = \mathbf{0}_{n \times n}.$$

2. **Proof of $A^\top A = \mathbf{0} \implies A = \mathbf{0}$**

Let $A^\top A = \mathbf{0}_{n \times n}$. For each $j \in \{1, 2, \dots, n\}$, the j -th diagonal entry of $A^\top A$ is computed as

$$\sum_{i=1}^m a_{ij} a_{ij} = \sum_{i=1}^m a_{ij}^2 = 0.$$

Since $a_{ij} \in \mathbb{R} \implies a_{ij}^2 \geq 0$ for all i , a sum of nonnegative real numbers equals zero if and only if each term is zero, which yields

$$a_{ij}^2 = 0 \implies a_{ij} = 0 \quad (\forall 1 \leq i \leq m, 1 \leq j \leq n) \implies A = \mathbf{0}_{m \times n}.$$

3. **Proof of $A = \mathbf{0} \implies AA^\top = \mathbf{0}$**

$$A = \mathbf{0} \implies a_{ij} = 0 \quad (\forall 1 \leq i \leq m, 1 \leq j \leq n).$$

The (i, k) -th entry of $AA^\top$ is given by

$$\sum_{j=1}^n a_{ij} a_{kj} = \sum_{j=1}^n 0 \cdot 0 = 0 \quad (\forall 1 \leq i, k \leq m) \implies AA^\top = \mathbf{0}_{m \times m}.$$

4. **Proof of $AA^\top = \mathbf{0} \implies A = \mathbf{0}$**

Let $AA^\top = \mathbf{0}_{m \times m}$. For each $i \in \{1, 2, \dots, m\}$, the i -th diagonal entry of $AA^\top$ is computed as

$$\sum_{j=1}^n a_{ij} a_{ij} = \sum_{j=1}^n a_{ij}^2 = 0.$$

Since $a_{ij} \in \mathbb{R} \implies a_{ij}^2 \geq 0$ for all j , a sum of nonnegative real numbers equals zero if and only if each term is zero, which yields

$$a_{ij}^2 = 0 \implies a_{ij} = 0 \quad (\forall 1 \leq i \leq m, 1 \leq j \leq n) \implies A = \mathbf{0}_{m \times n}.$$